

Sebastian Ibañez Rios

Software Engineer — Systems Engineering Student

Barranquilla, Colombia — sebdavidibanezrios@gmail.com — LinkedIn — GitHub — Portfolio

SUMMARY

Systems Engineering student and Junior Software Engineer with experience in backend development and biometric authentication systems. Proficient in modernizing legacy applications and building microservices using Java, Spring, AWS, OracleDB and DynamoDB. Proven ability to translate complex requirements into reliable solutions within Agile teams, combining strong technical foundations with clear communication and mentoring skills.

TECHNICAL SKILLS

- **Frontend:** React, Next.js, TypeScript, Tailwind CSS, Astro
- **Backend:** Python (FastAPI, Django), Node.js (NestJS, Express), Java (Spring Boot)
- **Databases:** PostgreSQL, MySQL, SQL Server, DynamoDB, Oracle
- **Tools & DevOps:** Git, GitHub, Docker, AWS, Vercel, Postman, Claude Code, Gemini CLI

PROFESSIONAL EXPERIENCE

Security Systems S.A.S

Aug 2025 – Present

Software Engineer

Barranquilla, Colombia (On-site)

- Maintain and evolve an enterprise-grade biometric authentication ecosystem for financial institutions, securely integrating industry-standard biometric solutions.
- Lead the architectural modernization of the main platform to current Java standards and enterprise application servers, significantly improving system performance and security.
- Design and implement scalable microservices utilizing Spring Boot, cloud-native containerized services, and NoSQL databases for highly available authentication.
- Collaborate on full-stack integration, developing modern web dashboards and desktop client applications to support the core infrastructure.

Fundación Rubato

Jul 2023 – Present

IT Coordinator & Music Technology Teacher

Barranquilla, Colombia (Hybrid)

- Managed internal systems and automated administrative workflows, significantly reducing manual tasks and optimizing the digital ecosystem.
- Teach professional notation (Finale, MuseScore) and lead sessions on AI and emerging technologies for music production.

PROJECTS

Orchestrator — *Python, Docker, API Gateway*

- Engineered a dynamic Docker microservices manager with an integrated API Gateway and ROBLE query authentication.

Climate_Node — *FastAPI, Streamlit, PostgreSQL*

- Built a statistical weather analysis engine for Colombia with automated cloud data fetching and PostgreSQL trend tracking.

WordFreq_MPI — *Python, MPI, Docker*

- Implemented a parallelized word frequency analysis tool that distributes processing across multiple nodes for large-scale data mining.

EDUCATION

Universidad del Norte

Expected Graduation: 2026

Systems Engineering Student

Barranquilla, Colombia

CERTIFICATIONS & COURSEWORK

DataCamp: Associate Data Engineer in SQL — **Coursework:** Database Design, Data Warehousing, Relational Databases, Cloud Computing, Data Engineering, Snowflake SQL.